

TENEBRIS
SYSTEMS

THE TEXTBOOK

Veloraxzia Tenebris

MATHEMATICS

GENERAL PLACEMENT EXAM

01.01.01.01.01

Unending; Unceasing; Eternal

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(1)

Algebra

(1.1)

Linear Equations and Inequalities

(1.1.1)

Solve for x in the following equation:

$$3(2x - 5) - 4(x + 1) = 7$$

(1.1.2)

Solve the following inequality and give the solution set in interval notation for the following equation:

$$-2(3x - 1) \leq 5x + 9$$

(1.1.3)

A phone plan charges a \$15 monthly fee plus \$0.08 per text. Write and solve an inequality for the number of texts t so the monthly cost is at most \$35.

(1.2)

The Coordinate Plane and Graphing

(1.2.1)

For the following line, find the slope and y -intercept of the line, then graph it:

$$2x - 3y = 12$$

(1.2.2)

Determine whether the points $(-2, 5)$, $(1, -1)$, and $(4, -7)$ are collinear.

(1.2.3)

A line passes through $(3, -2)$ and is perpendicular to the following line:

$$y = \frac{1}{3}x + 4$$

Find its equation in slope-intercept form.

(1.3)

Systems of Linear Equations and Inequalities

(1.3.1)

Solve the following system:

$$\begin{cases} 2x + 3y = 13 \\ x - 2y = -4 \end{cases}$$

(1.3.2)

Solve the following system:

$$\begin{cases} y \leq 3x - 1 \\ y = 5x - 6 \end{cases}$$

(1.3.3)

Graph the following system of inequalities and state the feasible region:

$$\begin{cases} y \geq -x + 1 \\ y < 2x + 4 \\ x \geq -1 \end{cases}$$

(1.4)

Exponents and Radicals

(1.4.1)

Simplify the following:

$$\frac{(x^3y^{-2})^2}{x^{-1}y^3}$$

(1.4.2)

Simplify and rationalize the denominator of the following:

$$\frac{5}{\sqrt{3} - \sqrt{2}}$$

(1.4.3)

Solve the following:

$$2^{x+1} = 16\sqrt{2}$$

(1.5)

Polynomials

(1.5.1)

Solve and simplify the following:

$$(2x - 3)(x^2 + 4x - 5)$$

(1.5.2)

Factor the following completely:

$$6x^3 - 15x^2 - 24x$$

(1.5.3)

Divide the following using synthetic division and state quotient and remainder:

$$(2x^3 - 3x^2 + 4x - 7) \div (x - 2)$$

(1.6)

Quadratics

(1.6.1)

Solve the following by completing the square

$$x^2 - 6x + 1 = 0$$

(1.6.2)

Find the vertex and axis of symmetry of the following:

$$y = -2x^2 + 8x - 3$$

State the maximum or minimum value.

(1.6.3)

A ball is thrown upward from a height of 5 m with initial velocity 18 m/s. Its height is given by the following:

$$h(t) = -4.9t^2 + 18t + 5$$

Find when it hits the ground.

(1.7)

Rational Expressions

(1.7.1)

Simplify the following and state the restrictions on x :

$$\frac{x^2 - 9}{x^2 - 3x}$$

(1.7.2)

Solve and check the following for extraneous solutions:

$$\frac{2}{x-1} + \frac{3}{x+1} = \frac{5}{x^2-1}$$

(1.7.3)

Find vertical and horizontal asymptotes of the following:

$$f(x) = \frac{3x^2 - 2}{x^2 - 4x + 4}$$

(1.8)

Absolute Value Equations and Inequalities

(1.8.1)

Solve the following:

$$|3x - 7| = 11$$

(1.8.2)

Solve the following inequality:

$$|2x + 1| < 5$$

(1.8.3)

A temperature sensor has an error of at most 0.3°C . If it reads 22°C , write an inequality for the true temperature T .

(1.9)

Functions

(1.9.1)

Given the following:

$$f(x) = \frac{x-1}{x+2}$$

Find the domain and range.

(1.9.2)

Let the following function definitions hold:

$$f(x) = x^2 - 4x + 3$$

$$g(x) = \sqrt{x+1}$$

Find the shared domain and range of both functions.

(1.9.3)

Determine whether the following equation is one-to-one:

$$f(x) = |x| + 2$$

Justify.

(2)

Precalculus

(2.1)

Function Families

(2.1.1)

Sketch the following functions:

$$f(x) = 2^x$$

$$g(x) = 2^{x-3} + 1$$

Describe transformations from f to g .

(2.1.2)

Find all intercepts and asymptotes of the following function:

$$f(x) = \frac{1}{x-2} + 3$$

(2.1.3)

A population doubles every 6 hours and starts at 500. Write the function for the population $P(t)$, and find $P(18)$.

(2.2)

Function Inverses and Compositions

(2.2.1)

Find the inverse of the following and verify by composition:

$$f(x) = \frac{3x-2}{5}$$

(2.2.2)

Given the following functions:

$$f(x) = \sqrt{4-x}$$

$$g(x) = x^2 + 1$$

Compute $(g \circ f)(x)$ and state its domain.

..... (2.2.3)

Determine whether the following is invertible on $\mathbb{R}$:

$$f(x) = x^3 - 3x$$

If not, give an interval where it is.

.....
..... (2.3)
.....
..... **Complex Numbers**
.....

..... (2.3.1)

Compute and write the following in $a + bi$ form:

$$(3 - 2i)(-1 + 5i)$$

..... (2.3.2)

Simplify the following:

$$\frac{4 + 7i}{2 - i}$$

..... (2.3.3)

Solve the following:

$$z^2 + 4z + 13 = 0$$

Express solutions in $a \pm bi$.

.....
..... (2.4)
..... **Zeros and Roots**
.....

..... (2.4.1)

Given the following function:

$$p(x) = x^3 - 6x^2 + 11x - 6$$

Find all real zeros and factor completely.

..... (2.4.2)

Use the Rational Root Theorem to list all possible rational zeros of the following function:

$$p(x) = 2x^3 - 3x^2 - 8x + 12$$

..... (2.4.3)

A polynomial has degree 4 with real coefficients and zeros $2, -1+3i$. Write a possible monic polynomial.

.....
..... (2.5)
..... Logarithms
.....

..... (2.5.1)

Solve the following:

$$\log_3(x-1) + \log_3(x+2) = 2$$

..... (2.5.2)

Rewrite the following in logarithmic form and evaluate:

$$5^{3/2}$$

..... (2.5.3)

Solve the following for t :

$$2e^{0.4t} = 15$$

.....
..... (2.6)
..... Sequences and Series
.....

..... (2.6.1)

Find an explicit formula for the arithmetic sequence with $a_1 = 7$ and common difference $d = -3$. Then find a_{20} .

..... (2.6.2)

Find the sum of the following geometric series:

$$\sum_{k=0}^8 3 \left(\frac{1}{2} \right)^k$$

..... (2.6.3)

Determine whether the following infinite geometric series converges, and if so find its sum:

$$1 - \frac{2}{3} + \frac{4}{9} - \dots$$

(2.7)

Combinatorics

(2.7.1)

How many 5-character passwords can be made using A–Z and 0–9 if repetition is allowed?

(2.7.2)

From 10 students, how many ways can you choose a committee of 4?

(2.7.3)

How many distinct permutations of the letters in MISSISSIPPI are there?

(2.8)

Probability

(2.8.1)

A fair die is rolled twice. Find $P(\text{sum} = 7)$.

(2.8.2)

A box has 5 red, 3 blue, and 2 green balls. Two are drawn without replacement. Find $P(\text{both red})$

(2.8.3)

Let the X be defined as following distribution:

$$X \sim \text{Binomial}(n = 12, p = 0.3)$$

Compute $P(X \geq 4)$.

(3)

Trigonometry

(3.1)

Radians and The Unit Circle

(3.1.1)

Convert 135° to radians and locate the angle on the unit circle.

(3.1.2)

Find the exact values of the following:

$$\sin\left(\frac{7\pi}{6}\right)$$

$$\cos\left(\frac{7\pi}{6}\right)$$

(3.1.3)

Given a point on the unit circle with $x = -\frac{3}{5}$ in Quadrant II, find y and $\tan(\theta)$.

(3.2)

Trigonometric Functions

(3.2.1)

Sketch one period of the following:

$$y = 3 \sin(2x - \pi)$$

State the amplitude, period, and phase shift.

(3.2.2)

Solve for $x \in [0, 2\pi)$ in the following:

$$\cos(x) = \frac{\sqrt{2}}{2}$$

(3.2.3)

Evaluate the following exactly:

$$\sec\left(\frac{\pi}{3}\right) + \csc\left(\frac{\pi}{6}\right)$$

(3.3)

Trigonometric Identities

(3.3.1)

Verify the following identity where it is defined:

$$\frac{1 - \cos 2x}{\sin 2x} = \tan(x)$$

(3.3.2)

Simplify the following:

$$\sin(x) \cos(x) (\tan(x) + \cot(x))$$

(3.3.3)

Prove the following identity:

$$\sin^2(x) = \frac{1 - \cos 2x}{2}$$

(3.4)

Vectors

(3.4.1)

Given the following:

$$\vec{\mathbf{a}} = \langle 2, -1, 3 \rangle$$

$$\vec{\mathbf{b}} = \langle -4, 0, 5 \rangle$$

Compute $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}$ and the angle between the vectors.

(3.4.2)

Find a unit vector in the following direction:

$$\vec{\mathbf{v}} = \langle 6, 2 \rangle$$

(3.4.3)

Compute the projection of $\vec{\mathbf{u}}$ onto $\vec{\mathbf{v}}$:

$$\vec{\mathbf{u}} = \langle 1, 2, 2 \rangle$$

$$\vec{\mathbf{v}} = \langle 2, 0, 1 \rangle$$

(3.5)

Polar Coordinates

(3.5.1)

Convert the following to Cartesian coordinates:

$$(r, \theta) = \left(4, \frac{5\pi}{6}\right)$$

(3.5.2)

Convert the following to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$:

$$(x, y) = (-3, 3\sqrt{3})$$

(3.5.3)

Sketch the polar curve for the following and identify it in Cartesian form:

$$r = 2 \cos(\theta)$$

(3.6)

The Complex Plane

(3.6.1)

Plot the following and compute the principal values $|z|$ and $\angle(z)$:

$$z = -2 + 3i$$

(3.6.2)

Write the following in $a + bi$ form:

$$z = 4 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right)$$

(3.6.3)

Compute using polar form:

$$\left(2 \operatorname{cis} \left(\frac{\pi}{6} \right) \right)^3$$

Where:

$$\operatorname{cis} \theta = \cos(\theta) + i \sin(\theta)$$

(4)

Calculus

(4.1)

Limits and Continuity

(4.1.1)

Compute the following:

$$\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right)$$

(4.1.2)

Determine whether the following is continuous at $x = 1$:

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x - 1, & x > 1 \end{cases}$$

(4.1.3)

Compute the following:

$$\lim_{x \rightarrow 0} \left(\frac{\sin(5x)}{x} \right)$$

(4.2)

Derivatives

(4.2.1)

Differentiate the following:

$$f(x) = x^3 \ln(x)$$

(4.2.2)

Find $f'(x)$ for the following:

$$f(x) = \frac{2x - 1}{x^2 + 4}$$

(4.2.3)

Find the derivative of the following:

$$y = e^{2x} \sin x$$

(4.3)

Implicit Differentiation

(4.3.1)

Given the following, find $\frac{dy}{dx}$:

$$x^2 + xy + y^2 = 7$$

(4.3.2)

Find the equation of the tangent line for the following at $(1, 0)$:

$$x^2y + \sin(y) = 1$$

(4.3.3)

Find $\frac{d^2y}{dx^2}$ for the following circle:

$$x^2 + y^2 = 25$$

(4.4)

Linear Approximations

(4.4.1)

Use linearization to approximate the following near $x = 8$:

$$\sqrt[3]{8.1}$$

(4.4.2)

Find the linearization of the following at $x = 1$, and use it to approximate $\ln(1.05)$:

$$f(x) = \ln(x)$$

(4.4.3)

Estimate the error of the next-order term for approximating $e^{0.02}$ using linearization at 0.

(4.5)

Related Rates

(4.5.1)

A circle's radius increases at at the following rate:

$$\frac{dr}{dt} = 0.4 \text{ m/s}$$

Find $\frac{dA}{dt}$ when $r = 5$ m.

(4.5.2)

A 10 m ladder slides down a wall. If the bottom moves away at 1.2 m/s , how fast is the top sliding when the bottom is 6 m from the wall?

(4.5.3)

Water is poured into a conical cup with a height of 12 cm and a radius of 4 cm. If $\frac{dV}{dt} = 3 \text{ cm}^3/\text{s}$, find $\frac{dh}{dt}$ when $h = 6$ cm.

(4.6)

Optimization

(4.6.1)

Find the dimensions of a rectangle with perimeter 40 that maximize area.

(4.6.2)

Minimize the surface area of a closed cylinder with a volume of 100π .

(4.6.3)

A company finds demand the following price per unit for x units:

$$p = 120 - 2x$$

Find the x that maximizes revenue.

(4.7)

Integrals

(4.7.1)

Compute the following:

$$\int 3x^2 - 4x + 1 dx$$

(4.7.2)

Compute the following:

$$\int_0^2 x^2 + 1 dx$$

(4.7.3)

Evaluate the following and state the domain condition:

$$\int \frac{1}{x} dx$$

(4.8)

Techniques of Integration

(4.8.1)

Compute the following:

$$\int x e^x dx$$

(4.8.2)

Compute the following:

$$\int \frac{1}{x^2 - 1} dx$$

(4.8.3)

Compute the following:

$$\int_0^{\pi/2} \sin^3(x) dx$$

(4.9)

Parameterization

(4.9.1)

Given the following, find $\frac{dy}{dx}$ in terms of t :

$$\vec{\mathbf{r}}(t) = \langle t^2, t^3 \rangle$$

(4.9.2)

Find the arc length of the following for $0 \leq t \leq \pi$:

$$\vec{\mathbf{r}}(t) = \langle 3 \cos(t), 3 \sin(t) \rangle$$

(4.9.3)

Eliminate the parameter to express the following curve as y in terms of x :

$$x = 2t - 1, y = t^2 + 4$$

(4.10)

Area and Average Value

(4.10.1)

Find the area between the following on $[0, 1]$:

$$y = x$$

$$y = x^2$$

(4.10.2)

Compute the average value of the following on $[0, 3]$:

$$f(x) = x^2$$

(4.10.3)

Set up the integral for the volume when the region under the following curve on $[0, 4]$ is revolved about the x -axis:

$$y = \sqrt{x}$$

(4.11)

Polar Calculus

(4.11.1)

Find the area enclosed by one loop of the following:

$$r = 2 \sin(\theta)$$

(4.11.2)

Compute the slope $\frac{dy}{dx}$ for the following polar curve at $\theta = \frac{\pi}{4}$:

$$r(\theta) = 1 + \cos(\theta)$$

(4.11.3)

Find the length of the following polar curve for $0 \leq \theta \leq 2\pi$:

$$r = 3$$

(4.12)

Convergence of Sequences and Series

(4.12.1)

Determine whether the following sum converges:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

(4.12.2)

Use the ratio test on the following:

$$\sum_{n=1}^{\infty} \frac{n!}{3^n}$$

(4.12.3)

Find the following:

$$\lim_{n \rightarrow \infty} \left(\left(1 + \frac{2}{n} \right)^n \right)$$

(4.13)

Power Series

(4.13.1)

Find the radius and interval of convergence of the following:

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n4^n}$$

(4.13.2)

Find the first four nonzero terms of the Maclaurin series for the following:

$$\sin(x)$$

(4.13.3)

Use a Taylor series about 0 to approximate the following with an error bound less than 10^{-4} :

$$\ln(1.2)$$